



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Multiband Compressor

Four-Band Dynamics

Four freely placed bands that compress or expand only where you need it, with an external sidechain, Mid/Side targeting, and zero-latency or linear-phase processing.

4 free dynamic bands

Compress or expand

Stereo, Mid, or Side

20 factory presets



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01 Overview

TB Multiband Compressor gives you four dynamic bands that you place anywhere in the spectrum. Unlike a crossover-based multiband, the bands do not split the whole signal: each band works only on its own range, and the rest of the spectrum passes untouched. Gaps and overlaps between bands are allowed.

Each band compresses or expands, and its signed **RANGE** sets the direction and the maximum change. A band can listen to its own range, to a separate detector range, or to an external sidechain, and it can work on the stereo signal, the Mid, or the Side. **PROCESSING** switches between a zero-latency Dynamic topology and a Linear-phase mastering topology.

Key features

- Four bands placed freely from 30 Hz to 18 kHz, 0.25 to 4 octaves wide
- Compression or expansion per band; the sign of RANGE selects downward or upward action
- Threshold, ratio, knee, attack, release, and band output per band
- Internal or shared external sidechain, with a band-linked or free detector range
- Stereo, Mid, or Side processing per band, with adjustable stereo link
- DYNAMIC (zero latency) or LINEAR phase, 2x to 8x oversampling, and 0 to 20 ms lookahead
- Interactive band graph with the input spectrum and Delta Listen
- 20 factory presets, user presets, undo, and redo

Signal flow



About this manual

This manual describes TB Multiband Compressor 1.0.2 for Windows (VST3). The current macOS build (AU and VST3) is version 1.0.1 and may differ.

02 Quick start

- 1 Insert TB Multiband Compressor on a track, bus, or the master. Init enables bands 1 to 3 at 120 Hz, 1.2 kHz, and 7 kHz with gentle compression; band 4 is off.
- 2 Select a band with the BAND buttons **4** or by clicking it in the graph **3**. Drag its node to set frequency and range, and drag a coloured edge to set the width.
- 3 Set **THRESHOLD**, **RANGE**, and **RATIO** **9**, then **ATTACK** and **RELEASE**.
- 4 Turn on **DELTA** **7** to hear only what the band changes, then turn it off again.
- 5 Set **MIX** **14** and compare with **BYPASS** **10**.

03 Interface



TB Multiband Compressor 1.0 interface with the Broad Control preset; band 1 is selected.

- 1 **Preset bar** — previous and next preset (hold to browse), the preset list, Save, Undo, and Redo.
- 2 **ANALYZER** — shows or hides the input spectrum.
- 3 **Band graph** — input spectrum, band nodes, coloured band edges, and the response.
- 4 **BAND 1–4, BAND ON** — selects a band and switches it on or off.
- 5 **COMP / EXPAND** — the band's dynamics mode.
- 6 **SIDECHAIN** — internal or external detector source.
- 7 **DELTA** — latches the band's wet-minus-dry monitor.
- 8 **Channel mode** — Stereo, Mid, or Side.
- 9 **Band controls** — FREQUENCY, WIDTH, THRESHOLD, RANGE, RATIO, KNEE, ATTACK, RELEASE.
- 10 **BYPASS** — global bypass.
- 11 **PROCESSING** — DYNAMIC or LINEAR.
- 12 **OVERSAMPLING** — OFF, 2X, 4X, or 8X.
- 13 **LOOKAHEAD / LAT** — lookahead time; the label shows the total latency in samples.
- 14 **MIX** — global dry/wet.

Working in the band graph

- Click a band to select it; double-click empty space to enable a band there.
- Drag a node to set the band's frequency and range.
- Drag either coloured band edge, or use the mouse wheel over a node, to set the width.
- Hold the right mouse button on a band node for momentary Delta Listen; drag sideways to move the band while listening, and release to stop.
- Double-click a value box to type an exact setting.

04 Controls

Band

Control	Range · default	What it does
BAND ON	Off, On · bands 1–3 on, band 4 off	Enables the selected band without changing the others.
FREQUENCY	30 Hz to 18 kHz · 120 Hz, 1.2 kHz, 7 kHz, 12 kHz	Centre frequency of the band.
WIDTH	0.25 to 4 oct · 1.5 oct	Width of the band in octaves.
THRESHOLD	–60 to 0 dB · –18 dB	Detector threshold.
RANGE	–48 to +12 dB · –3 dB	Signed maximum gain change. The sign selects the direction: negative values act downward, positive values upward.
RATIO	1:1 to 20:1 · 2:1	Compression or expansion ratio.
KNEE	0 to 18 dB · 6 dB	Soft-knee width.
ATTACK	0.5 to 200 ms · 15 ms	Detector attack time.
RELEASE	20 to 2000 ms · 180 ms	Detector release time.
COMP / EXPAND	Compress, Expand · Compress	Chooses compression or expansion for the band.
SIDECHAIN	Off, On · Off	Off uses the main input as the band's detector. On uses the host's shared auxiliary sidechain input; the panel shows NO SC INPUT when none is connected.
Channel mode	Stereo, Mid, Side · Stereo	Processes the stereo signal, the Mid, or the Side. Mono hosts fall back to Stereo.
DELTA	Off, On · Off	Latches the band's wet-minus-dry monitor, so you hear only what the band changes. It is not saved or automated.

Each band also has the following host parameters. They are stored in presets and the project, but they do not appear on the panel shown above.

Host parameter	Range · default	What it does
Band Output	−18 to +18 dB · 0 dB	Static output gain for the band.
Detect	BAND, FREE · BAND	BAND makes the detector follow the band's own range; FREE uses Detector Frequency and Detector Bandwidth.
Detector Frequency	30 Hz to 18 kHz · same as the band	Detector centre frequency in FREE mode.
Detector Bandwidth	0.25 to 4 oct · 1.5 oct	Detector width in FREE mode.
Stereo Link	0 to 100% · 100%	How strongly the two channels share one detector.
Audition	Off, On · Off	Band audition for monitoring.

Global

Control	Range · default	What it does
PROCESSING	DYNAMIC, LINEAR · DYNAMIC	DYNAMIC is a zero-latency band topology. LINEAR is a linear-phase mastering topology that adds FIR delay and can pre-ring.
OVERSAMPLING	OFF, 2X, 4X, 8X · OFF	Oversamples the dynamics path. Higher settings add latency and CPU use.
LOOKAHEAD	0 to 20 ms · 0 ms	Delays the gain action so the detector reacts ahead of the audio. It changes the reported latency.
MIX	0 to 100% · 100%	Blends the processed signal with the exact dry input.
Output (host parameter)	−18 to +18 dB · 0 dB	Final output gain.
BYPASS	Off, On · Off	Global bypass recognised by the host.

Latency

Only DYNAMIC with OVERSAMPLING OFF and LOOKAHEAD 0 ms has zero latency. The LOOKAHEAD / LAT label shows the current total latency in samples. PROCESSING, OVERSAMPLING, and LOOKAHEAD are not automatable, because changing them changes the reported latency.

05 Presets

Step through presets with the arrows (hold to browse), or open the list to choose a factory or user preset. Save stores the current settings as a .tbmbc user preset. Undo and Redo step through recent edits. In the table, the band list shows the active bands; PROCESSING gives the phase mode, oversampling, and lookahead.

Preset	Active bands	PROCESSING	MIX	Notes
START				
Init	120 Hz, 1.2 kHz, 7 kHz	DYNAMIC · OFF · 0 ms	100%	—
Gentle Balance	150 Hz, 1.2 kHz, 6.5 kHz	DYNAMIC · OFF · 0 ms	80%	—
Broad Control	90 Hz, 550 Hz, 2.8 kHz, 10.5 kHz	DYNAMIC · OFF · 0 ms	100%	—
SOURCE				
Vocal Focus	140 Hz, 1.8 kHz, 6.8 kHz	DYNAMIC · 2X · 2 ms	100%	Upward RANGE: 2; Mid: 1–3
Bass Foundation	70 Hz, 700 Hz	DYNAMIC · OFF · 0 ms	100%	Upward RANGE: 2; Mid: 1–2
Drum Tamer	80 Hz, 350 Hz, 3.2 kHz, 10 kHz	DYNAMIC · 4X · 3 ms	100%	—
Guitar Smooth	120 Hz, 2.5 kHz, 5.5 kHz, 10 kHz	DYNAMIC · 2X · 2 ms	100%	Mid: 1–2; Side: 4
Keys Space	90 Hz, 700 Hz, 3.5 kHz, 11 kHz	DYNAMIC · 4X · 1 ms	100%	Expand: 3–4; Upward RANGE: 3–4; Mid: 1, 3; Side: 4
BUS				
Mix Bus Balance	100 Hz, 700 Hz, 3.2 kHz, 11 kHz	DYNAMIC · 2X · 1 ms	90%	Mid: 1–2; Side: 4
Drum Bus Glue	90 Hz, 500 Hz, 3.5 kHz, 10.5 kHz	DYNAMIC · OFF · 0 ms	90%	—
Mastering Touch	90 Hz, 600 Hz, 3.5 kHz, 12 kHz	LINEAR · 2X · 5 ms	100%	Mid: 1; Side: 4
Parallel Drum Punch	75 Hz, 250 Hz, 3.2 kHz, 9.5 kHz	DYNAMIC · 8X · 1.5 ms	65%	Upward RANGE: 1, 3, 4; Mid: 1; Side: 4
MOTION				
Kick Duck	90 Hz	DYNAMIC · OFF · 3 ms	100%	Sidechain: 1
Rhythmic Pump	100 Hz, 800 Hz, 5 kHz	DYNAMIC · OFF · 0 ms	100%	Sidechain: 1–3
Transient Lift	100 Hz, 2.2 kHz, 8 kHz	DYNAMIC · 4X · 1 ms	100%	Upward RANGE: 1–3; Mid: 1–2; Side: 3
MASTER				
Transparent Master	70 Hz, 450 Hz, 3 kHz, 11 kHz	LINEAR · 4X · 5 ms	100%	Mid: 1–2; Side: 4
Loudness Guard	70 Hz, 350 Hz, 2.8 kHz, 8.5 kHz	DYNAMIC · 8X · 10 ms	100%	Mid: 1–2
De-Harsh Master	2.2 kHz, 4.2 kHz, 7.2 kHz, 12 kHz	LINEAR · 8X · 8 ms	100%	Mid: 1–2; Side: 3–4
Low End Anchor	48 Hz, 105 Hz, 250 Hz	LINEAR · 4X · 6 ms	100%	Mid: 1–2
Stereo Focus	90 Hz, 650 Hz, 3.5 kHz, 10.5 kHz	LINEAR · 4X · 5 ms	100%	Mid: 2; Side: 1, 4

Kick Duck and Rhythmic Pump use the external sidechain, so route a key signal, such as a kick drum, to the plug-in's sidechain input.

06 Recipes

Starting points for common jobs. Adjust them by ear.

Vocal control

Tame boom and harshness while keeping the vocal open.

Preset Vocal Focus

THRESHOLD -26 to -20 dB

- 1 Select the band over the problem range, narrow WIDTH, and use DELTA to hear what it removes.

Kick and bass

Duck the low end of a bass every time the kick hits.

Preset Kick Duck

SIDECHAIN On

- 1 Insert on the bass and route the kick to the sidechain input.

Gentle mastering

Transparent, linear-phase control of a full mix.

Preset Transparent Master

PROCESSING LINEAR

- 1 Keep RANGE small; a decibel or less per band is often enough.

Tighter stereo low end

Less low-frequency Side content on a mix.

Preset Stereo Focus

Channel mode Side
(band 1)

- 1 Listen in mono and in stereo as you set band 1's RANGE.

07 Tips and troubleshooting

Symptom	What to check
A sidechain preset does nothing.	No key signal reaches the plug-in. Route one to its sidechain input; the panel shows NO SC INPUT otherwise.
The band boosts instead of reducing.	RANGE is positive. Make it negative for downward action.
The track is late against others.	LINEAR, OVERSAMPLING, or LOOKAHEAD add latency. Check LOOKAHEAD / LAT and turn on delay compensation.
Transients get a soft pre-echo.	LINEAR phase can pre-ring. Use DYNAMIC on drums.
It is hard to hear what a band does.	Turn on DELTA, or hold the right mouse button on the band node.
Nothing changes.	The band is off (BAND ON), the signal stays below THRESHOLD, RANGE is 0 dB, or BYPASS is on.

08 Parameter reference

TB Multiband Compressor has 78 host parameters: six global ones and eighteen for each band. Band parameters are named Band 1 to Band 4.

Host parameter	Range	Default
GLOBAL		
Mix	0 to 100%	100%
Output	−18 to +18 dB	0 dB
Bypass	Off, On	Off
Phase Mode	Dynamic, Linear	Dynamic
Oversampling	OFF, 2x, 4x, 8x	OFF
Lookahead	0 to 20 ms	0 ms
EACH BAND (BAND 1 TO BAND 4)		
Enabled	Off, On	On for bands 1–3, off for band 4
Frequency	30 to 18000 Hz	120, 1200, 7000, 12000 Hz
Bandwidth	0.25 to 4 oct	1.5 oct
Threshold	−60 to 0 dB	−18 dB
Range	−48 to +12 dB	−3 dB
Ratio	1 to 20	2
Attack	0.5 to 200 ms	15 ms
Release	20 to 2000 ms	180 ms
Knee	0 to 18 dB	6 dB
Output	−18 to +18 dB	0 dB
Dynamics	Compress, Expand	Compress
Source	INT, EXT	INT
Detect	BAND, FREE	BAND
Detector Frequency	30 to 18000 Hz	Same as the band
Detector Bandwidth	0.25 to 4 oct	1.5 oct
Stereo Link	0 to 100%	100%
Audition	Off, On	Off
Channel Mode	Stereo, Mid, Side	Stereo

09 Version history

Version	Changes
1.0.2 · 2026-08-17	Stricter, portable numeric state parsing. All 78 parameter IDs, the state and preset formats, and the audio behaviour are unchanged.
1.0.1 · 2026-08-13	Response graph rework: RANGE now reads as the shaded base envelope, with the live gain reduction as an overlay tinted in the band colour; contours no longer draw over the 0 dB line.
1.0.0 · 2026-08-13	First release: four independently placed bands with compression or expansion, signed Range, external sidechain, stereo and Mid/Side targeting, Dynamic or Linear modes, oversampling, and lookahead.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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